Business Insights (On Power BI Dashboard)

Page 1 – Overview

- Total observations are high (~3K) with strong coverage across 99 locations and 79 species.
- Bird observations are heavily concentrated in grassland habitats compared to forests.
- Forest habitat is significantly underrepresented and may need more data collection.
- Monthly observations show a declining trend from May to July.
- Observation activity suggests possible seasonal or effort-based variation.
- A small number of observers contribute the majority of observations, creating concentration risk.
- A few species dominate the observation counts, indicating uneven species distribution.
- Visualization is the most used identification method, followed by singing and calling.
- Observations are recorded under moderate average temperature conditions (~23.78°C).
- Seasonal filters (Monsoon vs Summer) can reveal shifts in bird activity patterns.
- The dataset supports scalable biodiversity monitoring but needs better balance across habitats and observers.

Page 2 - Environmental Analysis

- Total observations are ~3K with average temperature ~24°C and humidity ~69%, indicating stable environmental conditions.
- Most observations occur under Partly Cloudy (1.6K ~46%) and Clear/Few Clouds (1.4K ~40%) conditions.
- Cloudy/Overcast conditions contribute significantly less (~0.5K ~14%), while Mist/Drizzle is negligible.
- 63% of observations occur during Monsoon, compared to 37% in Summer, showing strong seasonal dominance.
- Disturbances have minimal impact, with No effect (~1.7K ~49%) and Slight effect (~1.4K ~40%) dominating observations.
- Moderate and serious disturbances contribute very little (~0.3K and ~0.1K respectively, <12% combined).
- Majority of observations (~3.3K ~94%) occur when no flyover is observed, with only ~0.2K (~6%) during flyovers.

- Bird observations peak around 22–24°C (~380–364 counts), indicating optimal temperature range for activity.
- Observations gradually decline at extreme temperatures (<20°C or >30°C), dropping below ~100 counts.
- Environmental factors like sky condition, temperature, and disturbance collectively influence observation patterns, with clear and stable conditions driving the highest counts.

Page 3 - Conservation Insights

- Total watchlist observations are low (~22), indicating limited flagged species activity.
- Only 2 species are categorized as at-risk, showing a relatively small high-risk group.
- Stewardship status count is 15, indicating moderate conservation coverage.
- 75% of regions have no stewardship status, while only 25% are actively managed.
- Watchlist species presence is extremely low (~0K vs 3.4K non-watchlist), contributing to <1% of total observations.
- A few species dominate at-risk observations, with Field Sparrow (309) and Northern Cardinal (300) leading.
- Mid-tier at-risk species like Indigo Bunting (219) and Chipping Sparrow (213) show moderate presence.
- Lower-tier species such as Eastern Bluebird (106) and Eastern Towhee (109) have relatively fewer observations.
- At-risk species distribution is uneven, with top 2 species contributing a significant share of total counts.
- Overall biodiversity risk appears controlled but requires increased stewardship coverage and monitoring of key species.